

Chapter 78 — Anatomy of the Retroperitoneum — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Upper poles:** left 11th rib, right 12th; hila L1; lower poles L3–L4.
- **Supracostal access > subcostal complications;** supra-12th may cross pleura.
- **Intercostal bundle, top → bottom:** vein → artery → nerve; nerve least protected.
- **Perirenal space open inferiorly** → perinephric collections track to the pelvis.
- **Anterior lamina = Toldt; posterior lamina = Zuckerkandl** (thicker).
- **Marginal artery of Drummond** → IMA (inferior mesenteric artery) ligation safe.
- **Left gonadal vein → left renal vein at 90°** → left varicocele.
- **New right varicocele → image** for renal cell carcinoma with IVC (inferior vena cava) thrombus.
- **Left testis → left para-aortic; right testis → interaortocaval;** crossover only right → left.
- **Obturator nerve + lateral-wall TURBT** (transurethral resection of bladder tumour) → adduction → perforation.

DEFINITION, BOUNDARIES AND ORGANS

- **Boundaries:** peritoneum (front), abdominal wall (back), diaphragm (top), extraperitoneal pelvis (bottom).
- **Retroperitoneum ≠ extraperitoneal space** (latter also encircles the abdomen).
- **Primarily retroperitoneal:** adrenals, kidneys, ureters, aorta, IVC, anal canal.
- **Secondarily retroperitoneal:** ascending and descending colon, duodenum, pancreas.
- **Exceptions (intraperitoneal):** duodenum D1, pancreatic tail.
- **Lobstein 1829:** first used “retroperitoneal tumour”.
- **Gerota 1895:** anterior renal fascia. **Zuckerkindl 1883:** posterior.
- **Wickham 1979:** endoscopic ureteric stone removal in pneumoretroperitoneum.
- **Gaur 1992:** retroperitoneal videoscopic renal surgery.

BODY SURFACE LANDMARKS

- **Right kidney lower than left** (upper pole 12th vs 11th rib).
- **Hila L1; lower poles L3–L4.**

POSTERIOR ABDOMINAL WALL

- **External oblique:** ribs 5–12, fibres inferomedial; aponeurosis forms inguinal ligament.
- **Internal oblique:** fibres superomedial.
- **Order:** external oblique → internal oblique → transversus → transversalis fascia.
- **Psoas major:** T12–L5 → lesser trochanter; anterior rami L1–L3.
- **Psoas minor:** may be absent; inserts at pelvic brim.
- **Quadratus lumborum:** posterolateral to psoas; stabilises 12th rib.
- **Vertebrae:** 7 cervical, 12 thoracic, 5 lumbar.
- **Sacral cord segments at vertebral T12–L1;** specify vertebral vs segmental level.
- **Lower rib fracture** → suspect retroperitoneal injury.
- **11th rib: no neck; 12th rib: no angle,** shortest; both floating.
- **Intercostal bundle:** costal groove, lower border of rib above; between internal and innermost intercostals.
- **Superficial fascia:** Camper = fatty layer; Scarpa = membranous layer.
- **Costovertebral ligament** tethers 12th rib to L1–L2; divide it → more exposure.

LUMBODORSAL FASCIA

- **Three layers** fuse laterally; **access at 12th rib tip.**
- **Posterior layer:** covers erector spinae. **Middle layer:** erector spinae / quadratus lumborum. **Anterior layer:** front of quadratus lumborum.

- **Anterior layer** continuous with psoas fascia.
- **Dorsal lumbotomy:** vertical, lateral to erector spinae and quadratus lumborum; **no muscle cut.**

RETROPERITONEAL FASCIAE AND SPACES

- **Strata:** outer → transversalis; intermediate → renal fascia; inner → gut fusion fascia.
- **Posterior pararenal space:** posterior lamina (front); transversalis (back and side).
- **Transversalis–posterior lamina fusion:** incise to reach hilum.
- **Posterior lamina thicker,** seen more often on imaging.
- **Lateroconal fascia** separates anterior and posterior pararenal spaces.
- **Perirenal contents:** adrenal, kidney, ureter, fat, pedicle, gonadal vessels.
- **Perirenal fat fine, pale yellow vs pararenal coarse, yellow-orange.**
- **Perinephric collections:** urine, blood, pus; contained, but track caudally.
- **Perirenal spaces may communicate across** **midline** below hilum.
- **Anterior pararenal space:** colon, pancreas, **duodenum D2–D3.**
- **White line of Toldt:** lateral edge of colonic mesentery fusion; reflect colon here.

GASTROINTESTINAL VISCERA AND MESENTERY

- **Right renal surgery:** duodenum D2 at hilum → **Kocher manoeuvre;** pancreatic head over IVC.
- **Left renal surgery:** pancreatic tail, spleen, stomach at the upper pole.
- **Duodenum 20–25 cm; D3** between **SMA (superior mesenteric artery)** in front, aorta behind.
- **Ampulla of Vater:** common bile duct + pancreatic duct into D2.
- **Hepatocolic and splenocolic ligaments:** divide sharply; retraction tears liver and spleen.
- **Bogros space:** caudal extraperitoneal extension towards pelvis and bladder.
- **Close small mesenteric defects** after bowel isolation; internal hernia can be lethal.

VASCULATURE

- **Aorta enters T12; bifurcates L4. IVC exits T8.**
- **Levels:** celiac T12; SMA L1–L2; renal arteries L1–L2; IMA L3–L4.
- **Branch order:** inferior phrenic → celiac → middle adrenal → SMA → renal → gonadal → IMA.
- **Lumbar arteries:** usually **four pairs.**
- **Adrenal supply:** superior (inferior phrenic, **constant**), middle (aorta), inferior (renal).
- **Supernumerary renal arteries** ≥ 1 in 4; no side predominance.

- **Splenic vessels on cephalad pancreatic edge:** at risk lifting inferior pancreas off renal fascia.
- **SMA injury** in left surgery → severe bowel ischaemia despite collaterals.
- **Testicular artery → internal ring; ovarian artery → infundibulopelvic ligament.**
- **Safe to ligate:** IMA, gonadal arteries, median sacral. **Not safe:** SMA.
- **Ureteric supply:** abdominal ureter from medial; pelvic ureter from lateral.
- **Renal vein anterior to renal artery;** enters IVC at L1.
- **Right renal vein:** short, no tributaries, **duplicated in one-sixth.**
- **Left renal vein** receives gonadal, adrenal and ≥ 1 lumbar vein.
- **Right gonadal vein → IVC;** into right renal vein up to 10%.
- **Right adrenal vein:** short, single, into **posterolateral IVC.**
- **Right ureter:** dissect **lateral to gonadal vein;** medial plane → IVC injury.
- **IMV (inferior mesenteric vein) joins SMV (superior mesenteric vein) in 40%** (not splenic vein).
- **Median sacral vein → left common iliac;** avoid at sacrocolpopexy.
- **Ascending lumbar veins** behind psoas → azygos (right), hemiazygos (left).
- **Hepatic veins → anterior IVC** below diaphragm.
- **Budd-Chiari:** hepatic vein occlusion → jaundice, ascites, pain, hepatomegaly.

LYMPHATIC SYSTEM

- **Lymph flows cephalad, right → left → left innominate vein.**
- **Lumbar nodes drain lateral-branch organs:** kidney, adrenal, ureter, gonad.
- **Exception:** head, neck, right thorax, right arm, right heart → right innominate.
- **Left para-aortic:** aortic midline → left ureter. **Right paracaval:** IVC midline → right ureter.
- **Interaortocaval:** IVC midline → aortic midline.
- **Three testicular zones:** left para-aortic, interaortocaval, right paracaval.
- **Right testis** sends small but real drainage to left para-aortic.
- **Cisterna chyli:** behind right aorta, in front of **L1-L2** → thoracic duct.

NERVOUS STRUCTURES

- **Parasympathetic:** cranial nerves III, VII, IX, X + **S2-S4** (pelvic splanchnics).

- **Sympathetic:** cord segments **T1-L2;** thoracic, lumbar and sacral splanchnics.
- **Adrenal medulla** = modified ganglion; preganglionic synapse directly.
- **Pelvic splanchnics** = **only parasympathetic splanchnics.**
- **Sympathetic trunk medial to psoas;** lumbar vessels cross it at 90°.
- **Celiac plexus** first and largest; supplies kidney, adrenal, ureter.
- **Superior hypogastric plexus:** aortic bifurcation → L5.
- **Hypogastric plexus injury** → loss of emission / retrograde ejaculation.
- **Roots:** subcostal T12; iliohypogastric and ilioinguinal L1; genitofemoral L1-L2.
- **Roots:** lateral cutaneous nerve of thigh L2-L3; obturator and femoral L2-L4; sciatic L4-S3.
- **Upper plexus nerves pierce psoas;** lower nerves run medial to it.
- **Lateral cutaneous nerve of thigh:** purely sensory.
- **Pectineus:** dual supply, femoral + obturator.
- **Genitofemoral:** on psoas; motor to **cremaster;** injured in **psoas hitch, laparoscopic varicocelelectomy.**
- **Femoral:** retractor on inguinal ligament or low lithotomy → **no knee extension.**
- **Sciatic:** high lithotomy hyperflexion.

CLASSIC TRAPS

- **Trap:** right kidney higher — **Actually:** right lower (12th rib).
- **Trap:** all duodenum and pancreas retroperitoneal — **Actually:** not D1 or the tail.
- **Trap:** Gerota described both laminae — **Actually:** anterior only; posterior is Zuckerkandl.
- **Trap:** perirenal space closed below — **Actually:** open cone into pelvis.
- **Trap:** intercostal nerve protected in groove — **Actually:** lowest, often exposed.
- **Trap:** sympathetic outflow from vertebrae T1-L2 — **Actually:** cord segments T1-L2.
- **Trap:** IMV always joins splenic vein — **Actually:** joins SMV in 40%.
- **Trap:** vessel in reoperative left mesocolon is gonadal vein — **Actually:** may be IMV.
- **Trap:** left testis crosses to right paracaval — **Actually:** crossover only right → left.
- **Trap:** ilioinguinal is motor to cremaster — **Actually:** genital branch of genitofemoral.